BT-203 (GS)

B.Tech., I & II Semester Examination, June 2024

Grading System (GS)

BASIC MECHANICAL ENGINEERING

Time : Three Hours Maximum Marks : 70

Note : (i) Attempt any five questions. (ii) All questions carry equal marks. (iii) Original paper bilingual (Hindi + English) tha; yahan sirf English version diya gaya hai.

Coverage: Full paper (Q1–Q8)

1. (a) Draw Fe–Fe_3C phase diagram and explain the phase transformation reactions in the diagram. **7**

(b) Define the following mechanical properties of materials: (i) Strength (ii) Hardness (iii) Toughness (iv) Ductility (v) Brittleness (vi) Malleability. **7**

2. (a) Describe the typical shape of the stress–strain curve for a ductile material. Explain the significance of the yield point and ultimate tensile strength on the stress–strain curve. **7**

(b) State and explain various types of errors in measurements. **7**

3. (a) List out various operations performed by Lathe and explain some of them in detail. **7**

(b) List out various temperature measuring instruments and explain any two with neat diagram. **7**

4. (a) Describe the working principle of hydraulic pumps. Give classification of pumps with their applications. **7**

(b) Derive Bernoulli's equation for incompressible fluids. **7**

5. (a) What is Pascal's law? Explain how Pascal's law is applied in hydraulic systems. **7**

(b) What are the main types of boilers and how are they classified based on construction and operation? **7**

6. (a) Explain the concept of draught in boiler operation. Discuss the factors influencing the choice between natural and artificial draught systems in boiler installations. **7**

(b) Explain thermodynamic processes at constant pressure and volume. **7**

7. (a) With a neat sketch, explain the working of Carnot and derive the expression for thermal efficiency. **7**

(b) Explain the working principle of compressor with neat sketch. **7**

8. (a) Explain the working of a 2-stroke petrol engine with neat sketch. **7**

(b) Write short notes on Slip gauge, Sine-bar and Combination set. **7**

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